第4讲. 全微分. 可微条件.

作业: P38. 2(2). (5). (6). 4. 7.

P44. 1(2). 10(1).

$y=f(x)$. x_0. 可微.

$$f(x_0+\Delta x)-f(x_0)=A\Delta x+o(\Delta x) \quad (\Delta x\to 0)$$

定义1. ①若 $\lim\limits_{(x,y)\to(x_0,y_0)} f(x,y)=0$. 则 $f(x,y)$ 是当 $(x,y)\to(x_0,y_0)$ 时 无穷小量.

记子: $f(x,y)=o(1)$. $((x,y)\to(x_0,y_0))$

②. $f(x,y)=o(1)$. $g(x,y)=o(1)$. $((x,y)\to(x_0,y_0))$

1°. 若 $\exists\ \alpha,\beta>0$. s.t.

$$\alpha|g(x,y)|\le|f(x,y)|\le\beta|g(x,y)|$$

称 $f(x,y)$ 与 $g(x,y)$ 是 $(x,y)\to(x_0,y_0)$ 时同阶无穷小.

$$\sqrt{x^2+y^2}\le|x|+|y|\le\sqrt{2}\cdot\sqrt{x^2+y^2}$$

$(x,y)\to(0,0)$

$$\lim_{(x,y)\to(0,0)}\frac{f(x,y)}{\sqrt{x^2+y^2}}=0. \iff \lim_{(x,y)\to(0,0)}\frac{f(x,y)}{|x|+|y|}=0.$$

2°. $\rho=\sqrt{(x-x_0)^2+(y-y_0)^2}$. $(x,y)\to(x_0,y_0)$

$f(x,y)=o(1)$. $((x,y)\to(x_0,y_0))$

若 $\lim\limits_{\rho\to0}\dfrac{f(x,y)}{\rho^k}=0$, $k\in\mathbb{N}^+$.

则称 $f(x,y)$ 当 $(x,y)\to(x_0,y_0)$ 比 ρ^k 高阶无穷小量.

记 $f(x,y)=o(\rho^k)$. $((x,y)\to(x_0,y_0))$

定义2. 设 $z=f(x,y)$ 在 $P_0(x_0,y_0)$ 全P域内有定义. 若 $f(x,y)$ 在 P_0 处的全改变量

$$f(x_0+\Delta x,\ y_0+\Delta y)-f(x_0,y_0)$$

$$= \underline{A\Delta x + B\Delta y} + o(\rho) \qquad (\Delta x, \Delta y) \to (0,0)$$

$$\rho = \sqrt{\Delta x^2 + \Delta y^2}.$$

其中 A. B 与 Δx. Δy 无关常数

则 $f(x,y)$ 在 P_0 可微. 且 $A\Delta x + B\Delta y$ 称为 $f(x,y)$ 在 P_0 全微分.

记为 $df(x_0,y_0) = A\Delta x + B\Delta y$.

例1. 设 $f(x,y) = a_{11}x^2 + 2a_{12}xy + a_{22}y^2$.

证明: $f(x,y)$ 可微. 且求全微分.

$$\boxed{df(x_0,y_0) = A\,dx + B\,dy}$$

证明: $f(x,y) = a_{11}x^2 + 2a_{12}xy + a_{22}y^2 = (x, y)\underbrace{\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}}_{A}\begin{pmatrix} x \\ y \end{pmatrix}$

令 $X = (x, y)$ $\quad (x \to \Delta x,\ y \to \Delta y)$ $\quad = XAX^T = f(X)$

$\Delta X = (\Delta x, \Delta y)$

$$f(X+\Delta X) - f(X) = (X+\Delta X)A(X+\Delta X)^T - XAX^T$$

$$= \cancel{XAX^T} + XA(\Delta X)^T + \Delta XAX^T + \Delta XA(\Delta X)^T - \cancel{XAX^T}$$

$$= 2XA(\Delta X)^T + \Delta XA(\Delta X)^T$$

下证

$$\lim_{\|\Delta X\| \to 0} \frac{|\Delta XA(\Delta X)^T|}{\|\Delta X\|} = 0$$

$$f(X) = XAX^T. \qquad X = (x, y)$$

$\exists Q$. 满足 $Q^TQ = QQ^T = I$. s.t. $QAQ^T = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$

$XQ^T = Y = (y_1, y_2)$

$\underline{X = YQ}$. $\quad \|X\| = \|YQ\| = \|Q^TY^T\| = \sqrt{\langle Q^TY^T, Q^TY^T \rangle} = \sqrt{\langle QQ^TY^T, Y^T \rangle} = \|Y\|$.

$\min\{\lambda_1,\lambda_2\}\|Y\|^2 \le f(X) = XAX^T = YQAQ^TY^T = (y_1, y_2)\begin{pmatrix} \lambda_1 & \\ & \lambda_2 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$

$$= \lambda_1 y_1^2 + \lambda_2 y_2^2$$

$$\le \max(\lambda_1, \lambda_2) \cdot \|Y\|^2$$

$$\min\{\lambda_1,\lambda_2\}\|X\| \le \frac{f(X)}{\|X\|} \le \max\{\lambda_1,\lambda_2\}\|X\|$$

$\min\{\lambda_1,\lambda_2\}\|X\| \leq \frac{f(X)}{\|X\|} \leq \max\{\lambda_1,\lambda_2\}\|X\|$

$\therefore \lim\limits_{\|X\|\to 0} \frac{XAX^T}{\|X\|} = 0.$

$\therefore f(x,y)$ 可微. 且 $df(x,y) = 2XA(\Delta X)^T$

$= 2(x,y)\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22}\end{pmatrix}\begin{pmatrix}\Delta x \\ \Delta y\end{pmatrix}$

$= 2(a_{11}x+a_{12}y)\Delta x + 2(a_{12}x+a_{22}y)\Delta y$

定理1. 设 $z=f(x,y)$ 在 $P_0(x_0,y_0)$ 邻域内有定义.

若 $f(x,y)$ 在 P_0 可微. 且 $df(x_0,y_0) = A\Delta x + B\Delta y$ 则

① $f_x'(P_0)$, $f_y'(P_0)$ 存在. 且 $f_x'(P_0)=A$. $f_y'(P_0)=B$.

② $f(x,y)$ 在 P_0 连续.

证: $f(x_0+\Delta x, y_0+\Delta y) - f(x_0,y_0) = A\Delta x + B\Delta y + o(\rho)$ $(\rho\to 0)$

$\rho = \sqrt{\Delta x^2+\Delta y^2}$.

令 $\Delta y = 0$.

$\frac{f(x_0+\Delta x, y_0) - f(x_0,y_0)}{\Delta x} = A + \frac{o(\Delta x)}{\Delta x}$ $(\Delta x\to 0)$

$\downarrow$ $f_x'(P_0)$; $\frac{o(\Delta x)}{\Delta x} \to 0$

$f_y'(P_0) = B$.

$\lim\limits_{(\Delta x,\Delta y)\to(0,0)} f(x_0+\Delta x, y_0+\Delta y) = f(x_0,y_0)$

$\therefore f(x,y)$ 在 P_0 连续.

例2. 设 $f(x,y) = \begin{cases} y\arctan\frac{1}{\sqrt{x^2+y^2}} & (x,y)\neq(0,0) \\ 0 & (x,y)=(0,0)\end{cases}$

讨论 $f(x,y)$ 在 $(0,0)$ 可微性. 连续性.

分析. $\Delta f = f(\Delta x,\Delta y) - f(0,0)$

分析. $\Delta f = f(\Delta x, \Delta y) - f(0,0)$

$$\boxed{\text{若} \lim_{(\Delta x,\Delta y)\to(0,0)} \frac{\Delta f - f_x'(0,0)\Delta x - f_y'(0,0)\Delta y}{\sqrt{\Delta x^2 + \Delta y^2}} = 0 \quad \text{则 } f \text{ 在 } (0,0) \text{ 可微}}$$

$$f_x'(0,0) = \lim_{x\to 0} \frac{f(x,0) - f(0,0)}{x} = 0$$

$$f_y'(0,0) = \lim_{y\to 0} \frac{f(0,y) - f(0,0)}{y} = \lim_{y\to 0} \arctan\frac{1}{|y|} = \frac{\pi}{2}.$$

$$\lim_{(x,y)\to(0,0)} \frac{y\left(\arctan\frac{1}{\sqrt{x^2+y^2}} - \frac{\pi}{2}\right)}{\sqrt{x^2+y^2}} = 0$$

$$\left|\frac{y}{\sqrt{x^2+y^2}}\right| \le 1$$

$\therefore$ $f(x,y)$ 在 $(0,0)$ 可微，且连续.

例 3. 设 $f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$

　　讨论 $f(x,y)$ 在 $(0,0)$ 可微性. 偏导数

解: $f_x'(0,0) = \lim_{x\to 0} \dfrac{f(x,0) - f(0,0)}{x} = 0.$

$f_y'(0,0) = 0.$

$$\Delta f = f(x,y) - f(0,0) = \frac{xy}{\sqrt{x^2+y^2}}.$$

$$\lim_{\substack{(x,y)\to(0,0)\\ y=x}} \frac{\Delta f - f_x'(0,0)x - f_y'(0,0)y}{\sqrt{x^2+y^2}} = \lim_{\substack{(x,y)\to(0,0)\\ y=x}} \frac{xy}{x^2+y^2} = \frac{1}{2} \neq 0.$$

$\therefore$ $f(x,y)$ 在 $(0,0)$ 不可微.

定理 2. 设 $f(x,y)$ 在 $P_0(x_0,y_0)$ 邻域Ω内有定义.

$f(x,y)$ 在 $(0,0)$ 不可微.

定理2. 设 $f(x,y)$ 在 $P_0(x_0,y_0)$ 邻域 Ω 内有定义.

若 $\forall (x,y)\in\Omega$, $f_x'(x,y)$, $f_y'(x,y)$ 存在, 且在 P_0 连续.

则 $f(x,y)$ 在 P_0 可微, 且 $df(P_0) = f_x'(P_0)dx + f_y'(P_0)dy$.

证:
$$f(x_0+\Delta x, y_0+\Delta y) - f(x_0,y_0)$$
$$= \underline{f(x_0+\Delta x, y_0+\Delta y) - f(x_0, y_0+\Delta y)} + \underline{f(x_0, y_0+\Delta y) - f(x_0,y_0)}$$
$$\overset{\exists\theta_1\in(0,1)}{\underset{\exists\theta_2\in(0,1)}{=\!=}} f_x'(x_0+\theta_1\Delta x, y_0+\Delta y)\Delta x + f_y'(x_0, y_0+\theta_2\Delta y)\Delta y$$
$$= f_x'(x_0,y_0)\Delta x + f_y'(x_0,y_0)\Delta y + \alpha\Delta x + \beta\Delta y.$$

$f(x, y_0+\Delta y)$ $\quad$ x_0, $x_0+\Delta x$

$f(x_0, y)$ $\quad$ y_0, $y_0+\Delta y$

$$\because \lim_{(\Delta x,\Delta y)\to(0,0)} \Big(\underbrace{f_x'(x_0+\theta_1\Delta x, y_0+\Delta y) - f_x'(x_0,y_0)}_{\alpha}\Big) = 0$$

$$\lim_{(\Delta x,\Delta y)\to(0,0)} \Big(\underbrace{f_y'(x_0, y_0+\theta_2\Delta y) - f_y'(x_0,y_0)}_{\beta}\Big) = 0$$

下证:
$$\lim_{(\Delta x,\Delta y)\to(0,0)} \frac{\alpha\Delta x + \beta\Delta y}{\sqrt{\Delta x^2+\Delta y^2}} = 0$$

$$0 \le \left|\frac{\alpha\Delta x + \beta\Delta y}{\sqrt{\Delta x^2+\Delta y^2}}\right| \le |\alpha| \frac{|\Delta x|}{\sqrt{\Delta x^2+\Delta y^2}} + |\beta| \underbrace{\frac{|\Delta y|}{\sqrt{\Delta x^2+\Delta y^2}}}_{\le 1} \to 0$$

$(\Delta x, \Delta y)\to(0,0)$

$\therefore$ $f(x,y)$ 在 P_0 可微.

练习. 令 $f(x,y) = \begin{cases} (x^2+y^2)\sin\dfrac{1}{x^2+y^2} & (x,y)\ne(0,0) \\ 0 & (x,y)=(0,0) \end{cases}$

定理3. 设 $f(x,y)$ 在 $P_0(x_0,y_0)$ 邻域内有定义. 则 $f(x,y)$ 在 P_0 可微 $\Longleftrightarrow$

$$f(x_0+\Delta x, y_0+\Delta y) - f(x_0, y_0)$$

$$= f_x'(x_0, y_0)\Delta x + f_y'(x_0, y_0)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y.$$

$$\lim_{(\Delta x, \Delta y)\to(0,0)} \varepsilon_i = 0. \quad (i=1,2)$$

证："$\Rightarrow$"。设 $f(x,y)$ 在 P_0 可微。

$$f(x_0+\Delta x, y_0+\Delta y) - f(x_0, y_0)$$

$$= f_x'(x_0, y_0)\Delta x + f_y'(x_0, y_0)\Delta y + \alpha \quad (\rho\to 0)$$

$$\rho = \sqrt{\Delta x^2+\Delta y^2}.$$

其中

$$\lim_{(\Delta x, \Delta y)\to(0,0)} \frac{\alpha}{\sqrt{\Delta x^2+\Delta y^2}} = 0.$$

$$\alpha = \underbrace{\alpha\frac{\operatorname{sgn}\Delta x}{|\Delta x|+|\Delta y|}}_{=\varepsilon_1}\Delta x + \underbrace{\alpha\frac{\operatorname{sgn}\Delta y}{|\Delta x|+|\Delta y|}}_{=\varepsilon_2}\Delta y$$

$$\Downarrow$$

$$\lim_{(\Delta x, \Delta y)\to(0,0)} \frac{\alpha}{|\Delta x|+|\Delta y|} = 0.$$

2. 求函数近似值。

$$f(x_0+\Delta x, y_0+\Delta x) \approx f(x_0, y_0) + f_x'(x_0, y_0)\Delta x + f_y'(x_0, y_0)\Delta y$$

$$(\rho\to 0)$$

$$\rho = \sqrt{\Delta x^2+\Delta y^2}$$

例. 求 $\sqrt{3.01^2+3.98^2}$ 近似值.

$(x_0, y_0) = (3, 4)$ $\quad \Delta x = 0.01, \quad \Delta y = -0.02$

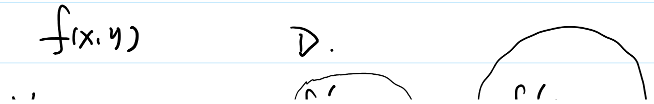
$f(x,y) = \sqrt{x^2+y^2}$. $\quad f_x' = \dfrac{x}{\sqrt{x^2+y^2}} \quad f_y' = \dfrac{y}{\sqrt{x^2+y^2}}$

$$\sqrt{3.01^2+3.98^2} \approx f(3,4) + f_x'(3,4)\Delta x + f_y'(3,4)\Delta y$$

$$= 5 + \frac{3}{5}\cdot 0.01 + \frac{4}{5}(-0.02) = 4.99.$$

3.

$f(x,y)$ $\qquad$ D.



$f(x,y)$ D.

$\forall (x,y) \in D$. $f_x'(x,y)$ $f_y'(x,y)$

$\forall P_0(x_0,y_0) \in D$.

$f_x'(x,y_0)$ $x=x_0$ 求导.

若 $\lim\limits_{\Delta x \to 0} \dfrac{f_x'(x_0+\Delta x, y_0) - f_x'(x_0,y_0)}{\Delta x}$ 存在.

$\|$

$f_{xx}''(x_0,y_0)$ —— $f(x,y)$ 在 (x_0,y_0) 关于 x 2阶偏导数.

$$\left.\frac{\partial\left(\frac{\partial f}{\partial x}\right)}{\partial x}\right|_{(x_0,y_0)} = \left.\frac{\partial^2 f}{\partial x^2}\right|_{(x_0,y_0)} \text{ 或 } \frac{\partial^2 f}{\partial x^2}(x_0,y_0)$$

$f_x'(x_0,y)$ 在 $y=y_0$ 求导.

若 $\lim\limits_{\Delta y \to 0} \dfrac{f_x'(x_0, y_0+\Delta y) - f_x'(x_0,y_0)}{\Delta y}$ 存在

$\|$

$f_{xy}''(x_0,y_0)$ —— $f(x,y)$ 在 (x_0,y_0) 先对 x 后对 y 二阶混合偏导数.

$$\left.\frac{\partial\left(\frac{\partial f}{\partial x}\right)}{\partial y}\right|_{(x_0,y_0)} = \left.\frac{\partial^2 f}{\partial y\,\partial x}\right|_{(x_0,y_0)}$$

$f_y'(x,y_0)$ 在 $x=x_0$ 求导

若 $\lim\limits_{\Delta x \to 0} \dfrac{f_y'(x_0+\Delta x, y_0) - f_y'(x_0,y_0)}{\Delta x}$ 存在.

$\|$

$f_{yx}''(P_0)$ —— $f(x,y)$ 在 P_0 先对 y 后对 x 二阶混合偏导数.

$$\left.\frac{\partial\left(\frac{\partial f}{\partial y}\right)}{\partial x}\right|_{P_0} = \left.\frac{\partial^2 f}{\partial x\,\partial y}\right|_{P_0}$$

$f_y'(x_0,y)$

若 $\lim \dfrac{f_y'(x_0, y_0+\Delta y) - f_y'(x_0,y_0)}{\quad}$ 存在

若 $\lim\limits_{\Delta y\to 0}\dfrac{f_y'(x_0,y_0+\Delta y)-f_y'(x_0,y_0)}{\Delta y}$ 存在

$\parallel$

$f_{yy}''(P_0)$ $\quad$ $f(x,y)$ 在 P_0 对 y 二阶偏导数

$$\frac{\partial\left(\frac{\partial f}{\partial y}\right)}{\partial y}\Big|_{P_0} = \frac{\partial^2 f}{\partial y^2}\Big|_{P_0}. \quad \frac{\partial^2 f}{\partial y^2}(P_0)$$

例. 设 $f(x,y)=\begin{cases} xy\dfrac{x^2-y^2}{x^2+y^2} & (x,y)\neq(0,0) \\ 0 & (x,y)=(0,0)\end{cases}$

求 $f_{xy}''(0,0)$, $f_{yx}''(0,0)$.

解: $f_{xy}''(0,0)=\lim\limits_{y\to 0}\dfrac{f_x'(0,y)-f_x'(0,0)}{y}=-1.$

$y\neq 0$, $f_x'(0,y)=\lim\limits_{x\to 0}\dfrac{f(x,y)-f(0,y)}{x}=\lim\limits_{x\to 0}y\dfrac{x^2-y^2}{x^2+y^2}=-y.$

$f_x'(0,0)=\lim\limits_{x\to 0}\dfrac{f(x,0)-f(0,0)}{x}=0$

$f_{yx}''(0,0)=\lim\limits_{x\to 0}\dfrac{f_y'(x,0)-f_y'(0,0)}{x}=1.$

$\forall x\neq 0$. $f_y'(x,0)=\lim\limits_{y\to 0}\dfrac{f(x,y)-f(x,0)}{y}=\lim\limits_{y\to 0}x\dfrac{x^2-y^2}{x^2+y^2}=x.$

$f_y'(0,0)=0.$

定理4. 设 $f(x,y)$ 在 $P_0(x_0,y_0)$ 邻域 Ω 内存在 f_{xy}'', f_{yx}'', 且均在 P_0 连续.

则 $f_{xy}''(P_0)=f_{yx}''(P_0)$.

证: Δx, Δy.

任固定 $\Delta y\neq 0$

$$\varphi(x)=f(x,y_0+\Delta y)-f(x,y_0)$$

$$\varphi'(x)=f_x'(x,y_0+\Delta y)-f_x'(x,y_0)$$

$$\Delta=\varphi(x_0+\Delta x)-\varphi(x_0)=\varphi'(x_0+\theta_1\Delta x)\Delta x \quad (\theta_1\in(0,1))$$

$\varphi(x_0+\Delta x)-\varphi(x_0) = f(x_0+\Delta x,y_0+\Delta y)-f(x_0+\Delta x,y_0)-f(x_0,y_0+\Delta y)+f(x_0,y_0)$

$$=\left(f_x'(x_0+\theta_1\Delta x,y_0+\Delta y)-f_x'(x_0+\theta_1\Delta x,y_0)\right)\Delta x$$

$\exists\theta_2\in(0,1)$

$$= \left(f_x'(x_0+\theta_1\Delta x,\ y_0+\Delta y) - f_x'(x_0+\theta_1\Delta x,\ y_0) \right) \Delta x$$

$$\overset{\exists\theta_2\in(0,1)}{=\!=} f''_{xy}(x_0+\theta_1\Delta x,\ y_0+\theta_2\Delta y)\,\Delta x\Delta y$$

$\Delta x \neq 0$.

$$\psi(y) = f(x_0+\Delta x,\ y) - f(x_0,\ y)$$

$$\psi'(y) = f_y'(x_0+\Delta x,\ y) - f_y'(x_0,\ y)$$

$$\Delta = \psi(y_0+\Delta y) - \psi(y_0) = \psi'(y_0+\theta_3\Delta y)\,\Delta y \qquad (\theta_3\in(0,1))$$

$$= \left(f_y'(x_0+\Delta x,\ y_0+\theta_3\Delta y) - f_y'(x_0,\ y_0+\theta_3\Delta y) \right) \Delta y$$

$$\overset{\exists\theta_4\in(0,1)}{=\!=} f''_{yx}(x_0+\theta_4\Delta x,\ y_0+\theta_3\Delta y)\,\Delta x\Delta y$$

$$\therefore\quad f''_{xy}(P_0) = f''_{yx}(P_0).$$